Meal photo + EXIF
day, band, zoom

Capture gates
invalid input abstains

Scale anchor
60 mm ArUco on A4

SAM 2.1
select one food mask

SigLIP 2
classify one dish

accepted dish

unsupported

cannot_verify
invalid / unrecognized / error

MoGe-2 depth
marker calibrates metric scale

out_of_scope
unsupported dish (e.g., wheat)

Scale tier?

no marker

WED mapped?

measured

mapped

unmapped

Measured marker
geometry mass: area × height × density

Head available?
mapped dish only

Unmapped dish
size-prior geometry

yes

no

Measured grams
from calibrated geometry

Learned mass
frozen DINOv2-S head

cannot_verify
mapped head absent or fails

Prior grams
coverage ×0.60

Nutrient uncertainty
Monte Carlo, 4,000 samples

Compliance policy
Energy/protein PASS: all P ≥ 0.90, C ≥ 0.85
FAIL: any P ≤ 0.10, C ≥ 0.90

PASS

BORDERLINE

FAIL

Serialized response
verdict, reasons, failures